

Threat Intelligence & MITRE ATT&CK Mapping

Overview

This portfolio report presents a concise threat-intelligence comparison of APT1, APT38, and APT41 based on MITRE ATT&CK analysis and defender-focused interpretation. The project compares each group's objectives, representative tactics and techniques, and operational patterns to show how structured intelligence analysis can support hunting, detection, and incident-response planning.

Project scope

The analysis focused on three advanced persistent threat groups with different strategic goals:

- APT1: primarily associated with long-term espionage and intellectual-property theft.
- APT38: primarily associated with financially motivated intrusions, including activity linked to SWIFT-related theft.
- APT41: associated with both espionage and financially motivated operations, often combining broad access techniques with flexible command-and-control behavior.

The report used MITRE ATT&CK as the main analytical framework and compared behaviors across reconnaissance, initial access, execution, persistence, lateral movement, command and control, exfiltration, and impact.

Threat group comparison

Group	Primary objective	Representative techniques and behaviors	Defender takeaway
APT1	Economic espionage and theft of sensitive business information	Spearphishing, custom backdoors, use of valid accounts, encrypted command-and-control traffic, long-term collection and exfiltration	Defenders should monitor for phishing-driven access, unusual authentication behavior, sustained internal discovery, and staged exfiltration activity.
APT38	Financial theft and disruptive operations against banking environments	Long dwell time, exploitation of public-facing systems, credential theft, customized malware, destructive actions, SWIFT-focused targeting	Defenders should prioritize network segmentation, credential protection, monitoring around high-value financial systems, and response playbooks for destructive malware or fraud activity.
APT41	Mixed espionage and financial gain	Supply chain compromise, spearphishing, exploitation of public-facing applications, lateral movement through remote services, adaptive command and control	Defenders should hunt for web-shell activity, credential abuse, remote administration misuse, and blended intrusion patterns that cross espionage and financially motivated behaviors.

Key observations

Several patterns stood out across the three groups:

- All three actors rely on adaptable access methods rather than one single entry technique.
- Credential abuse, internal discovery, and command-and-control concealment remain recurring themes across the groups.
- The biggest differences appear in mission focus: APT1 emphasizes espionage, APT38 emphasizes financial theft and operational disruption, and APT41 combines espionage with financially motivated activity.
- MITRE ATT&CK mapping helps turn descriptive threat reporting into specific detection and hunting priorities.

Defensive recommendations

The analysis supports several practical defensive actions:

1. Improve phishing resistance through awareness training, email controls, and attachment or link inspection.
2. Strengthen credential security with multi-factor authentication, privileged account monitoring, and credential hygiene.
3. Use ATT&CK-aligned detection logic to monitor for discovery, lateral movement, suspicious remote access, and covert command-and-control behaviors.
4. Protect critical systems through network segmentation, logging, and targeted monitoring for sensitive financial or administrative assets.
5. Maintain incident-response playbooks and tested backup or recovery procedures to reduce the impact of destructive or financially motivated attacks.

Portfolio value

This project demonstrates the ability to compare adversary behavior across multiple threat groups, translate ATT&CK analysis into defender language, and connect intelligence findings to practical security operations outcomes. It also shows structured reporting, framework-based analysis, and the ability to convert technical patterns into clear security recommendations.